THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 10: **How Quantities Combine: Understanding Data** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A group of 24 students has an average of 68 marks and another group of 16 students has an average of 78 marks. The average of all 40 students is: 1

(a) 74 **(b)** 71 **(c)** 73 **(d)** 72

2. A group of 18 students has an average of 45 marks and another group of 12 students has an average of 70 marks. The average of all 30 students is: 1

(a) 54 **(b)** 56 **(c)** 57 **(d)** 55

3. The averages of 4 months of 30 days each are 7, 9, 14, 10. The overall average is: 1

(a) 9 **(b)** 11 **(c)** 12 **(d)** 10

4. 2 litres of a solution with 4% salt is mixed with 3 litres of a solution with 9% salt. The concentration of salt in the mixture is: 1

(a) 7% **(b)** 8% **(c)** 6% **(d)** 9%

5. 200 units of a mixture with 10% metal is mixed with a mixture of 25% metal to get 15% metal. The quantity of the second mixture needed is (same units): 1

(a) 120 **(b)** 80 **(c)** 200 **(d)** 100

6. Marks of 90, 70 are combined with weights 3 : 2. The weighted average is: 1

(a) 83 **(b)** 81 **(c)** 82 **(d)** 84

7. The average of 15 values is 62. When one more value, 78, is added, the new average is: 1

(a) 63 **(b)** 64 **(c)** 65 **(d)** 62

8. How much water must be added to 20 litres of a 6% solution to reduce its concentration to 4%? 1

(a) 20 litres **(b)** 12 litres **(c)** 10 litres **(d)** 11 litres

9. Brahmagupta used a weighted-mean formula to find the: 1

(a) interest on a loan **(b)** mean depth of an irregularly shaped pit
(c) volume of a cone **(d)** area of a circle

10. A 100% stacked bar chart is best when we want to compare: 1

(a) only the totals of the bars
(b) exact values of the parts
(c) the order of the bars alone
(d) the proportions of the parts within each bar

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The average marks of 30 girls are 70 and of 20 boys are 60. Find the average of all 50 students. 2

12. A cyclist rides 5 days each in Week 1 and Week 2 with weekly averages of 24 min and 30 min. Find the average over the 10 days. 2

13. 200 mL of a 10% sugar solution is mixed with 300 mL of a 20% sugar solution. Find the concentration of the mixture. 2

14. Marks of 80, 60 and 70 are combined in the ratio 2 : 1 : 1. Find the weighted average. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The average weight of 7 adults is 70 kg and of 3 children is 40 kg. Explain why $\frac{70+40}{2}$ is not the average of all 10 persons, and find the correct value. 3

16. A school has 40 students in Class A averaging 72 marks and 60 in Class B averaging 80. Find the combined average and say whether it is nearer to 72 or 80, with a reason. 3

17. Alloy A (150 kg) has 60% copper and Alloy B (250 kg) has 40% copper. Find the percentage of copper when they are melted together. 3

OR

How much of a 30% solution must be added to 10 litres of a 60% solution to get a 50% solution? 3

18. Sana's marks are 58% in internals, 66% in the project and 75% in the final exam, combined in the ratio 3 : 2 : 5. Find her annual score and compare with the simple average. 3

OR

Tara scores 40/50 in internals, 45/60 in the project and 72/80 in the final exam, with weights 2 : 3 : 5. Find her percentage. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A cricket academy has 9 seniors with average height 168 cm and 6 juniors with average height 152 cm. 5

(i) Find the sum of the heights of the seniors and of the juniors.

(ii) Find the average height of all 15.

(iii) Why does $\frac{168+152}{2}$ give a wrong result?

OR

(i) Prove that the combined average of collections of n and m values with averages a and b is $\frac{an+bm}{n+m}$. 5

(ii) Hence show that when $n = m$ it reduces to $\frac{a+b}{2}$.

20. **Case study:** Alloy P weighs 100 kg with 40% silver, Alloy Q weighs 150 kg with 20% silver, and Alloy R has 60% silver. When all three are melted together, the new alloy has 40% silver. 5

(i) Write the equation for the weight y of Alloy R.

(ii) Solve it.

(iii) Verify by finding the silver in the final alloy.

OR

Case study: At a stall, the pani has 8% spice. Customers complain. The owner wants to reduce the spice to $\frac{3}{4}$ of the original concentration. 5

(i) What is the new concentration?

(ii) How much water must be added to 10 litres?

(iii) What is the total volume after adding?

END OF QUESTION PAPER — **Total: 40 Marks**